

NTUST course: Computer Vision and Applications (CI5336701, 2026 Spring)

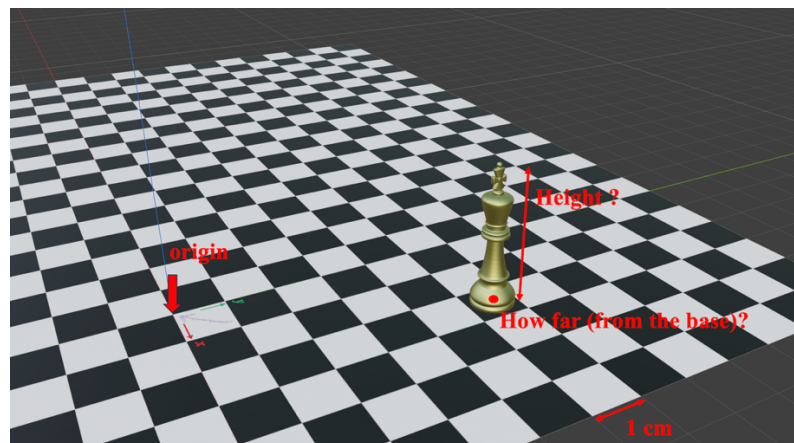
Homework#3: Estimate how far and the height of the chess in one image.

Date Due : 2026 May 12th, PM11:59 (around 2.0 weeks)

Description

1. The goal of this assignment is to implement a perspective projection model to estimate the physical dimensions and spatial position of a 3D object from a single 2D image. Specifically, you are required to determine the height of the golden chess piece and its distance from the camera center. One PNG image with known intrinsic parameter is given.
2. In the figure, the size of each checker on the ground is 1cm x 1cm. The origin of world coordinate is also shown for your reference.

K matrix =[1866.7, 0.00, 960.0
0.00, 1866.7, 540.0
0.00, 0.00, 1.0]



3. In this homework, you can use any method including third-party lib. to perform, but please do NOT directly use any commercial software to create the image for this assignment.
4. Deliverable: there are **at least two types** of data you should upload to <https://moodle2.ntust.edu.tw> by date due
 - 1) Source code in Python, C++/C or Matlab. (including simple comments to show the result)
 - 2) Report (~two pages):
 - a. Explain how you calculate / estimate the values based on any assumption (better to proof);
 - b. all markers you refer in the image should be disclosed in the report.
 - 3) Optional: Execution files (.exe and related dynamic link file .dll).
5. Grade evaluation
 - **Mathematical assumption:** Correct application of the perspective projection model and intrinsic parameters (50%).

- **Accuracy of Distance Estimation:** Proximity of the calculated distance to the ground truth (30%).
- **Accuracy of Height Estimation:** Proximity of the calculated height to the ground truth (20%).

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